Prosodic Phrases as Planning Units in Spoken Language

During language production, speakers must manage the competing demands of effective communication and cognitive efficiency. This trade-off is particularly relevant to theories of prosodic phrasing, which often posit that prosody conveys structural elements of language. Such accounts of prosody reflect an architecture whereby prosody is used to bridge multiple levels of linguistic representation, but place less emphasis on domain-general cognitive constraints on language production such as the need to continually refresh working memory during speech planning [1]. If prosodic phrasing is driven at least in part by speech planning, then the relation between prosody and syntactic structure may be a consequence of planning scope — typically individual phrases [2,3] — rather than a reflection of linguistic structure.

Because studies of prosody in language production typically make use of grammatically structured sequences to analyze prosodic features, it is difficult to determine the unique contribution of planning scope in prosodic phrasing. However, evidence from typed recall of numerical lists demonstrates that response timing systematically increases at regular positions, providing evidence that spontaneous chunking in short-term memory impacts the temporal structure of the output [4]. In the current research, we employ a similar approach using verbal recall to specifically to set aside those structural properties that are thought to affect or even determine prosodic phrasing in order to assess the extent to which prosodic phrases are related to speech planning, as indicated by chunking behavior in short-term memory. In other words, what prosodic phrasing would people apply to an unstructured list, deprived of any syntactic or other type of compositional structure?

57 undergraduate participants completed an auditory serial recall task in which they were asked to listen to 10 numeric lists that included the English numbers 1-9 in a random order. The lists were creating using text-to-speech, with a one-second pause in between each item. Participants were told to listen carefully to each list, and then orally repeat the items back in the order that they heard them as accurately as possible. Trials that included speech errors and filled pauses were excluded from further analysis. Recordings were force-aligned to identify onset and offset times of each item, and prosodically analyzed using the Wavelet Prosody Toolkit [5] for the presence of acoustic features associated with prosodic boundaries: word lengthening, intensity, and pitch variations.

Figure 1 shows the distribution of prosodic boundary strengths. A two-gamma mixture model was used to identify strong prosodic boundaries. Recall accuracy was measured using Levenstein's distance, and a median split was used to divide the data based on high- and low-accuracy trials to examine how task performance impacted placement of prosodic boundaries. Results were analyzed using ANOVAs, with planned contrasts comparing boundary strength and pause durations at recall positions 3 and 6 to all other serial positions.

As seen in Figure 2, the average prosodic boundary strength for all words shows a significant increase at recall positions 3 and 6, but only when recall was more accurate (ps < .01). Further, greater recall accuracy resulted in more prosodic boundaries at the third and sixth items in the list (Figure 3). In contrast, when recall was lower in accuracy, placement of prosodic boundaries occurred more often in recall positions 2-4, suggesting that cognitive efficiency during recall influences the size of prosodic units. Finally, the time between the onset of a word representing a prosodic boundary to the onset of the next prosodic phrase was longer overall when recall accuracy was lower ($p < .001$), suggesting that difficulty during recall results in greater pauses following prosodic phrases to allow time to retrieve the next item (Figure 4).

These results suggest that, in the absence of linguistic structure, speakers generate prosodic phrases that reflect units of recall from working memory, revealing some of the ways resource availability during speech planning may shape the prosodic features of speech. Overall, these results carry important implications for the role of speech planning and phonological working memory store in prosodic phrasing during spoken language, and highlight how non-linguistic factors influence prosodic structure.

Figure 1. Distribution of prosodic boundary strengths as calculated by the Prosody Wavelet Toolkit [5]. A two-gamma mixture model was used to group strong and weak prosodic boundaries, and a cutoff value was calculated by identifying the lowest value that was predicted to belong to the distribution representing strong prosodic boundaries (red line).

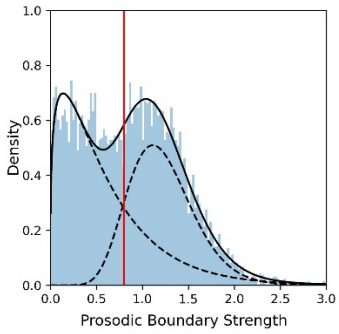


Figure 2. Average prosodic boundary strength at each serial position, excluding list-final items.

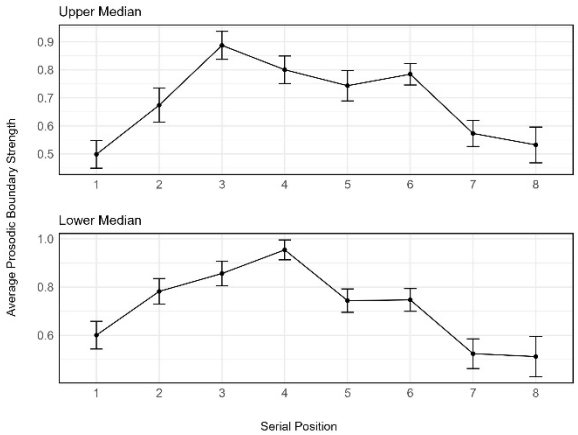


Figure 3. Average number of items in each prosodic phrase for high (top panel) and low (bottom panel) recall accuracy. Numbers in parentheses following each serial position on the x-axis indicate the number of prosodic boundaries identified at that position. The black dots reflect the overall average, and grey dots reflect individual participant averages.

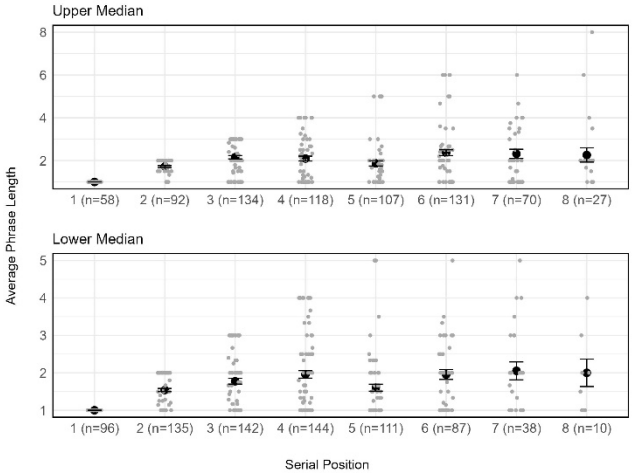
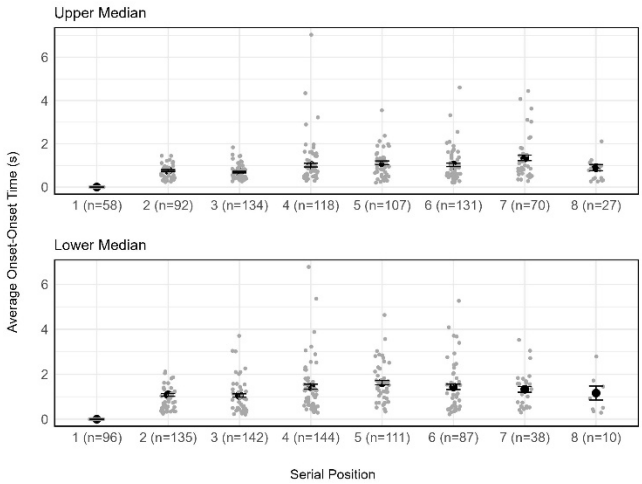


Figure 4. Average time between the onset of a word representing a prosodic boundary to the onset of the next prosodic phrase high (top panel) and low (bottom panel) recall accuracy. Numbers in parentheses following each serial position on the x-axis indicate the number of prosodic boundaries identified at that position. The black dots reflect the overall average, and grey dots reflect individual participant averages.



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